FREQUENCY-DOMAIN ANALYSIS BASED EXPLOITATION OF COLOR CHANNELS FOR COLOR IMAGE DEMOSAICKING

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ABSTRACT

Color-difference interpolation (CDI) has been a widely used technique for various color demosaicking methods. CDI-based methods perform interpolation in the color-difference domain assuming that the color-difference signal is a low-pass signal. Recently, a residual interpolation (RI) algorithm, which conducts interpolation in the residual domain, has been developed, and it assumes that the residual domain is flatter or smoother than the channel-difference domain. In this paper, we comprehensively show a frequency domain analysis of these assumptions and observe that it is image dependent and creates artifacts in the interpolated image. With this view, we propose an algorithm that uses the inter-color correlation as well as the residual smoothness among the different channel much better than the existing algorithms. Experimental results emphasize that the proposed algorithm atribute better performances the existing algorithms in terms of both visual and objective quality.

Index Terms— Color-difference interpolation, demosaicking, color filter array (CFA), prediction, frequency domain.

1. INTRODUCTION

Generally, color images consists of three bands and they are red ($\mathbf{R}$), blue ($\mathbf{B}$), and green ($\mathbf{G}$). To comprehensively perceive the image, it is essential to have all three charge-coupled devices to capture three components, which is a economically expensive. In order to further decrease the cost of production of most digital cameras, a single image sensor is overlaid with a color filter array (CFA). The CFA such as Bayer pattern (shown in Fig. 1) allows only one color to be measured at each pixel. To restore a full-color image, the other two missing color elements at each pixel are estimated and is known as color demosaicking.

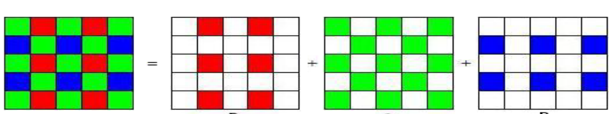

Fig. 1. Bayer pattern, US Patent 3971065.

Recently, various demosaicking algorithms have been proposed in the literature. Since sampling frequency of the $\mathbf{G}$ band is more than the $\mathbf{R}$ and $\mathbf{B}$ channels (as shown in Fig. 1), so nearly all of the CFA demosaicking algorithms first predict the green channel and then further use this information to predict the remaining channels. For reconstruction of other bands, most of the demosaicking algorithms use color correlation among the bands. The color-difference interpolation (CDI) based approach [2-6] is extensively used CFA demosaicking algorithms that uses inter-color correlation between the different channels. The rationale behind channel-correlation is based on the assumption that different color channels share similar high-frequency structures [2-6] and so channel-difference ($\mathbf{B}$-$\mathbf{G}$ or $\mathbf{R}$-$\mathbf{G}$) information can be considered to be as low-pass signals.

Recently, residual interpolation (RI) based methods [7-9] were proposed to achieve a superior performance over CDI algorithms. RI-based algorithms first get an approximate image of the image with missing pixels via a guided filter [11] and then estimate residual image between the original and the tentative image at the available pixels. They assume that the residual image is smoother than a color difference image and thus apply a low-pass filter to reconstruct the undersampled components. We observe, however, that the assumptions made by the CDI- and RI-based algorithms are image dependent and thus may not hold for all images.

In this paper, we comprehenively show the frequency domain analysis of these assumptions and explain what happens when a low-pass filter is applied. We observe, however, that both the assumptions are image dependent and subsequently, degrading the overall pecceptual quality. To address this issue, an efficient algorithm is proposed that uses the inter-color correlation characteristics as well as the residual information of the bands effectively and subsequently achieves better performance.

2. REVIEW OF CDI AND RI-BASED APPROACHES

Both the CDI- and RI-based methods focus on the generation of the $\mathbf{B}$ and $\mathbf{R}$ band, as they are significantly undersampled as compared to $\mathbf{G}$ band. A working process of these methods is given as below.

2.1. Review of CDI-based approaches

CDI-based approaches first reconstructs the green channel as number of available pixels in $\mathbf{G}$ are more than $\mathbf{R}$ or $\mathbf{B}$ channels. The predicted green component is denoted as $\hat{\mathbf{G}}$. Then the CDI scheme estimates the subsampled channel-difference signal $K_R = R_a - \hat{G}_a$ at the available $\mathbf{R}$ pixels, where R_a and $\hat{G}_a$ denote the original red ($\mathbf{R}$) and reconstructed $\mathbf{G}$ components sampled at the available $\mathbf{R}$ positions. CDI assumes that the color-difference signal ($K = \mathbf{R} - \hat{\mathbf{G}}$) is a low-pass signal [1] and the interpolated red component ($\hat{R}_{CDI}$) using CDI algorithm is given in (1),

$$\hat{R}_{CDI} = \zeta\{K_R\} + \hat{G} = \zeta\{R_a - \hat{G}_a\} + \hat{G}, \qquad (1)$$

where $\zeta\{.\}$ represent the low-pass filter (LPF) is applied to the subsampled channel-difference image (K_R) and later predicted $\hat{G}$ is added to get the predicted R component. Furthermore CDI assumes

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that the LPF of R_a and $\hat{G}_a$ gives exactly the low frequencies of $\mathbf{R}$ (R_l) and $\hat{\mathbf{G}}$ ($\hat{G}_l$), and interpolated $\hat{R}_{CDI}$ from (1) can be written as

$$\hat{R}_{CDI} = \zeta\{R_a - \hat{G}_a\} + \hat{G} = (R_l - \hat{G}_l) + \hat{G}. \tag{2}$$

Using the similar procedure, the blue channel can be predicted.

2.2. Review of RI-based approaches

RI-based approaches [7-9] have been recently developed to achieve the bettter results as compared to the color-difference interpolation (CDI) methods. The working procedure of RI-based schemes is as follows. Like CDI schemes, RI-based algorithms focus on the under-sampled components, such as $\mathbf{R}$ and $\mathbf{B}$. They first estimate a approximate image (denoted as $[.]^T$) of an image with missing pixels via a guided filter [11] and assumes that such filter can preserve the edges in the approximate image. Then they estimate a subsampled residual image $Res_R = R_a - R_a^T$ at the available $\mathbf{R}$ pixels, where R_a and R_a^T denote the original red ($\mathbf{R}$) and the approximate image (R^T) at available $\mathbf{R}$ positions, as given in Fig.1. RI-based approaches assume that the residual image ($Res = \mathbf{R} - \mathbf{R^T}$) is smoother than the channel-difference signal ($K = \mathbf{R} - \hat{\mathbf{G}}$), and they apply an LPF on the subsampled residual image to get the reconstructed red component ($\hat{R}_{RI}$), which is given in (3):

$$\begin{aligned} \hat{R}_{RI} = \zeta\{Res_R\} + R^T = \zeta\{R_a - R_a^T\} + R^T \\ \implies \hat{R}_{RI} = (R_l - R_l^T) + R^T, \end{aligned} \tag{3}$$

where $\zeta\{.\}$ represents LPF. Like CDI, they assume that LPF of R_a and R_a^T gives very similar the low frequencies of $\mathbf{R}$ (R_l) and R^T (R_l^T). Then approximate image (R^T) is merged to the residual image in (3) for the reconstruction.

3. PROPOSED ALGORITHM

CDI-based approaches assume that the channel-difference image ($K = \mathbf{R} - \hat{\mathbf{G}}$) is a low-pass signal because of high inter-color correlation, whereas RI-based approaches assume that the residual image ($Res = \mathbf{R} - \mathbf{R^T}$) are more flatter than the color-difference image because of the guided filter. We observe that the assumptions made by these approaches are not valid for all the datasets, and hence they produce artifacts which reduce the quality of the reconstructed image. To solve this issue, we comprehensively show a frequency domain analysis of these assumptions, and based on the observation, we propose an efficient demosaicking algorithm which works for various kind of images.

3.1. Frequency domain analysis of assumptions

Let $\mathbf{O}$ is the pristine high-resolution input image of dimension $\mathbf{M} \times \mathbf{N}$, here $\mathbf{O} \in \{\mathbf{R}, \mathbf{G}, \mathbf{B}\}$. While $\mathbf{O_{CFA}}$ is the undersampled image generated using the respective CFA pattern to the pristine image. For instance, $\mathbf{R_{CFA}}$ is generated by applying the respective CFA pattern to the pristinel $\mathbf{R}$ image. A 2×2 patch of $\mathbf{R_{CFA}}$ will have one red component, as given in Fig. 1, and the missing color components in $\mathbf{R_{CFA}}$ are shown by zero. The R_{CFA} can be generated from $\mathbf{R}$ as

$$R_{CFA}(i,j) = \begin{cases} R(i,j), & \text{for } \left(i = 2k-1, \text{ j=}2l\right) \\ 0 & \text{otherwise}, \end{cases} \tag{4}$$

where $k = 1, ..., M/2$, $l = 1, ..., N/2$ and $i = 1, ..., M$, $j = 1, ..., N$. For the analysis of the assumption of the RI method in (3), subsampled residual image $Res_R = R_{CFA} - R^T$ is predicted at the available $\mathbf{R}$ pixels and is shown as

$$Res_R(i,j) = \begin{cases} R_{CFA}(i,j) - R^T(i,j), \text{for } \left(i = 2k-1, \text{ j=}2l\right) \\ 0 \qquad \text{otherwise}. \end{cases} \tag{5}$$

Equation (5) can be further simplified as

$$Res_R(i,j) = (R(i,j) - R^T(i,j))\Delta^R_{CFA}(i,j), \tag{6}$$

where $\Delta^R_{CFA}(i,j)$ is a modulation signal, and it is represented as

$$\Delta^R_{CFA}(i,j) = \frac{1}{4}(1-(-1)^i)(1+(-1)^j), \tag{7}$$

The Discrete-time Fourier transform (DTFT) of (7) can be shown by

$$\tilde{\Delta}^R_{CFA}(u,v) = \frac{1}{4}.\mathbf{\Delta(u)}' \begin{bmatrix} -1 & -1 & -1 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} .\mathbf{\Delta(v)}, \tag{8}$$

where the DTFT is denoted as $\tilde{[.]}$, $\Delta(u) = [\delta(u+\frac{1}{2}), \delta(u), \delta(u-\frac{1}{2})]'$, $\Delta(v) = [\delta(v+\frac{1}{2}), \delta(v), \delta(v-\frac{1}{2})]'$, and $[.]'$ denotes transposition. Taking the DTFT of (6) and using (8), we get,

$$\tilde{Res}_R(u,v) = \frac{1}{4}. \mathbf{1_3}'. \begin{bmatrix} -\tilde{Res}(u^+,v) & -\tilde{Res}(u^+,v) & -\tilde{Res}(u^+,v^-) \\ \tilde{Res}(u,v^+) & \tilde{Res}(u,v) & \tilde{Res}(u,v^-) \\ -\tilde{Res}(u^-,v^+) & -\tilde{Res}(u^-,v) & -\tilde{Res}(u^-,v^-) \end{bmatrix} .\mathbf{1_3}, \tag{9}$$

where $\tilde{Res}(u,v) = \tilde{R}(u,v) - \tilde{R}^T(u,v))$ is the DTFT of the residual image ($Res = \mathbf{R} - \mathbf{R^T}$), $(u^{\pm}, v^{\pm}) = (u \pm \frac{1}{2}, v \pm \frac{1}{2})$, and $1_3 = [1 \;\; 1 \;\; 1]'$. We have evaluated the magnitude of the subsampled residual image ($\tilde{Res}_R(u,v)$) in (9), and it is shown in Fig. 2(a). In Fig. 2(a), one can observe, there are nine replicated spectra of $\tilde{Res}(u,v)$. By definition in (3), RI-based approaches apply an LPF to the subsampled residual image (Res_R), which implies that in the frequency domain the LPF will extract the center spectrum ($\tilde{Res}(u,v)$) from the nine spectra.

Similarly if we study the assumption of the subsampled channel-difference signal (K_R) in (1) and follow a similar derivation to that given above, we can write the DTFT of K_R as

$$\tilde{K}_R(u,v) = \frac{1}{4}. \mathbf{1_3}'. \begin{bmatrix} -\tilde{K}(u^+,v) & -\tilde{K}(u^+,v) & -\tilde{K}(u^+,v^-) \\ \tilde{K}(u,v^+) & \tilde{K}(u,v) & \tilde{K}(u,v^-) \\ -\tilde{K}(u^-,v^+) & -\tilde{K}(u^-,v) & -\tilde{K}(u^-,v^-) \end{bmatrix} .\mathbf{1_3}, \tag{10}$$

where $\tilde{K}(u,v)$ is the DTFT of the channel-difference signal ($K = \mathbf{R} - \hat{\mathbf{G}}$). We have evaluated the magnitude of $\tilde{K}_R(u,v)$ in (10), and it is shown in Fig. 2(b). Like RI methods, CDI methods also apply an LPF to the subsampled channel-difference signal in (1), which also implies that in the frequency domain the LPF will extract the center spectrum from the nine spectra. In Fig. 2, for both cases, we can see that the center spectrum may have interference from the surrounding spectra and applying an LPF will extract the center spectrum with some additional artifacts.

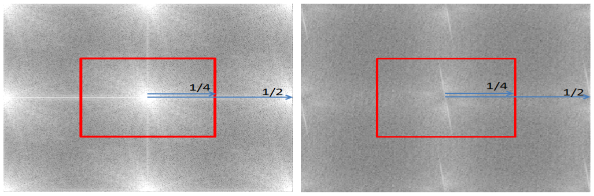

Fig. 2. Frequency spectra of subsampled (a) residual image (Res_R) and (b) color-difference image (K_R) with a low-pass filter.

3.2. The proposed algorithm for demosaicking

With the above shown, frequency domain analysis, we understand that both the CDI and RI-based approaches depend on the energy of the center spectrum. RI algorithms assume that the residual image (Res) is smoother or flatter than the color-difference image (K), i.e, the frequency spectrum of the residual image (center spectrum in Fig. 2(a)) is more compact than the frequency spectrum of the color-difference image (center spectrum in Fig. 2(b)) and thus applying an LPF to the subsampled residual image will extract the center spectrum with less additional artifacts. We emphasize that this assumption is highly image dependent and does not stand true for images having high inter-color correlation. In such cases, the color-difference image is smoother than residual image. With this view, we propose an algorithm which linearly combines both the approaches on the basis of patch-by-patch to have a better prediction for demosaicking. We denote the proposed predicted red component by $\hat{R}_{PRO}$, and which is shown as:

$$\hat{r}_{PRO} = w_1 \hat{r}_{RI} + w_2 \hat{r}_{CDI}. \tag{11}$$

Here $\hat{r}_{PRO} \in \hat{R}_{PRO}$, $\hat{r}_{RI} \in \hat{R}_{RI}$ and $\hat{r}_{CDI} \in \hat{R}_{CDI}$ are the interpolated patches of size $m \times n \in M \times N$ obtained using the proposed algorithm, RI algorithm and CDI method. w_1 and w_2 are the weights to combine such that $w_1 + w_2 = 1$. We propose two different schemes to generate the weights (w_1, w_2), and these are explained in the following.

3.2.1. Scheme 1 : Weight estimation based on the spectrum energy

In this scheme, we estimate the weights (w_1, w_2) based on the energy of the center spectrum. For each approach, we calculate the spectrum energy by squaring each frequency within the LPF (Fig. 2) and sum them up to get

$$SE_{Res} = \sum_{(u,v)\in(F_c)} (\tilde{Res}_R(u,v))^2, SE_K = \sum_{(u,v)\in(F_c)} (\tilde{K}_R(u,v))^2. \tag{12}$$

Here, SE_{Res} and SE_K are the spectrum energy of the center spectrum of the subsampled residual image (Fig. 2(a)) and color-difference image (fig. 2(b)) respectively within the LPF. It is known from (9) and (10) that the frequency spectra are symmetric (Fig. 2), and thus we keep the cut-off frequency (F_c) of the LPF as half of the distance between two spectrum, i.e., $F_c = 1/4$ for both cases.

In (12), the spectrum energy with the smallest value indicates that the corresponding approach in Fig. 2 will extract the center spectrum with minimum interference and thus a better reconstruction is possible. For example, if $SE_K < SE_{Res}$, then intuitively, for better reconstruction, the weight of r_{CDI} should be more in (11) as compared to the other method, i.e, $w_1 < w_2$. With the arguments given above, we propose to find the weights by taking the reciprocal of the various spectrum energy values in (12) and normalizing it as follows:

$$w_1 = (SE_{Res})^{-1}/N, w_2 = (SE_K)^{-1}/N), \tag{13}$$

where $N = (SE_{Res})^{-1} + (SE_K)^{-1}$, is the normalization factor.

3.2.2. Scheme 2 :Weight estimation based on linear minimum mean square sense (LMMSE)

In this scheme, we estimate the optimal weights, and this is explained as follows. We can write the distortion (d) between the predicted block ($r \in \mathbf{R}$) and original block ($\hat{r}_{PRO}$) as

$$d = r - \hat{r}_{PRO} = r - (w_1 \hat{r}_{RI} + w_2 \hat{r}_{CDI}) = w_1 d_{RI} + w_2 d_{CDI}. \tag{14}$$

Here, $d_{RI} \in D_{RI}$ and $d_{CDI} \in D_{CDI}$ are the distorted blocks estimated using the corresponding RI and CDI algorithms. To get the LMMSE optimal weights (w_1, w_2), problem can be formulated as below:

$$\min_{w_1, w_2} \quad E[d^2], \quad \text{s.t.} \quad \sum_{i=1,2} w_i = 1. \tag{15}$$

The optimal weights are estimated by differentiating above objective function with respect to w_1 and w_2

$$\begin{cases} w_1 = E[d_{CDI}(d_{CDI} - d_{RI})]/E[(d_{CDI} - d_{RI})^2], \\ w_2 = E[d_{RI}(d_{RI} - d_{CDI})]/E[(d_{CDI} - d_{RI})^2]. \end{cases} \tag{16}$$

So with the substitution of (16) into (11), we can efficiently predict the red component's unknown pixels in a patch. To get d_{RI} and d_{CDI} in (16), the original patch (r) is required, which can not be available in practical scenarios. With this view, we treat resultant image of the method [9] as an approximation of the original block (r). However, any other algorithm can also be used for the same purpose.

4. EXPERIMENTAL RESULTS

We compared the performance of the proposed algorithm with the state-of-the-art demosaicking algorithms for Bayer pattern. For the experimental results, we used two standard datasets, IMAX [4] and KODAK. The IMAX dataset contains 18 images with the size of 500×500, and the KODAK dataset contains 12 images of size 768 $\times 512$. compared the performance of the proposed algorithm with several methods including CDI-[4] and RI-[7-9] based interpolation.

The proposed algorithm works on a patch-by-patch basis to have a better prediction for demosaicking and we kept the block size at 6×6 for both datasets. At the same time, the patch size can be modified depending upon the image resolution. Since RI and CDI algorithms focus more on the reconstruction of the $\mathbf{R}$ and $\mathbf{B}$ band, we only show the results for these two bands. In order to get the $\hat{\mathbf{G}}$ component, we have used algorithm proposed in [9]. We use simple averaging filter as LPF (for equation (2) and (3)) in our simulations.

Table 1 gives the average PSNR performance of the proposed algorithm on the two datasets. For the IMAX dataset, we can see that the proposed algorithm respectively using the two proposed weight (PRO1, PRO2) outperforms the existing algorithms in terms of average PSNR. Our algorithm also outperforms the RI-based algorithms [7-9] and CDI-based algorithms on the KODAK dataset.

In Fig. 3, we show the visual analysis of the weight (w_2) map for the reconstruction of the red component of a test image, where proposed weight map1 and proposed weight map2 refer to the proposed algorithm with weight scheme 1 and with weight scheme 2

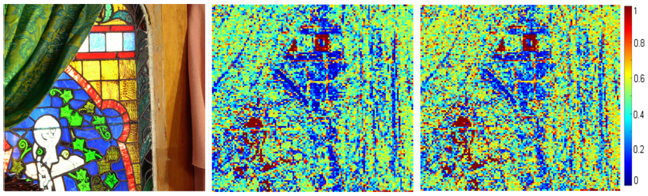

Fig. 3. Weight map (w_2) analysis: (a) Test image, (b) Proposed weight map1, (c) Proposed weight map2.

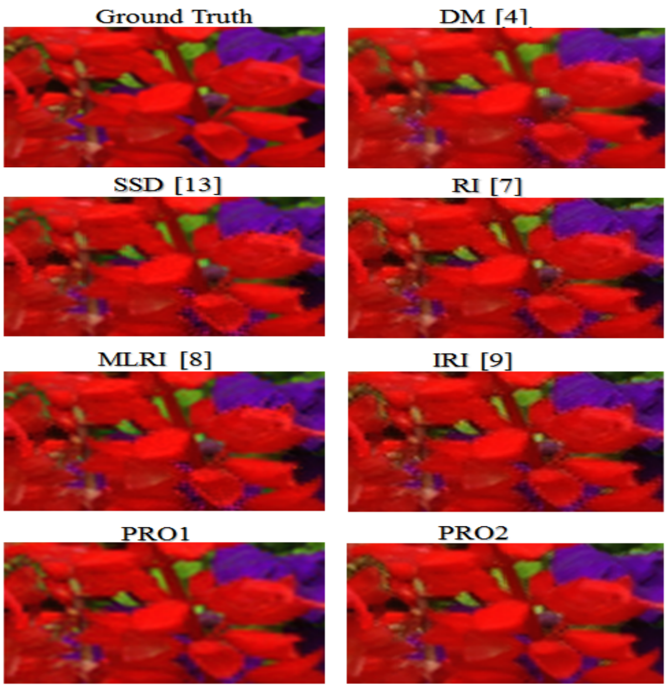

Fig. 4. (a) Subjective comparison of different demosaicking algorithms on a test image (IMAX). Here, PRO1, PRO2 represents to weighting scheme 1 and 2 integrated with the proposed algorithm.

respectively. In Fig. 3(a) and 3(b), red dots denote that $w_2 = 1$, which implies that the CDI method is given higher weightage in (11), whereas blue dots denote $w_2 = 0$ and $w_1 = 1$, which implies that the RI method is given higher weightage in (11) for the reconstruction. It is clear that both the weight maps don't have much variation, and thus the performance of both schemes, as shown in Table 1, are similar. To show the effectiveness of the proposed algorithm, visual comparison of reconstructed results of different algorithms are displayed in Fig. 4. From Fig. 4, it is clear that the existing algorithms create artifacts on the red leaves, on the other hand, the proposed algorithm achieves superior results.

5. CONCLUSION

In this work, we have presented an efficient CFA demosaicking algorithm which uses the inter-color correlation as well as the residual-smoothness property of the color channels. We have undertaken a frequency domain analysis to study the characteristics of the channel-difference signal, and residual-difference signal, and based on our observation, we have proposed an algorithm that generates

Table 1. Average PSNR results on the IMAX and KODAK dataset. Here, PRO1, PRO2 represents to weighting scheme 1 and 2 integrated with the proposed algorithm.

Algo.	IMAX		KODAK		IMAX+KODAK	
	R	B	R	B	R	B
SSD [13]	35.02	33.80	38.83	39.08	36.54	35.91
NAT [4]	36.31	34.50	38.30	37.94	37.11	35.87
ECC [14]	36.67	35.31	39.87	39.00	37.95	36.77
RI [7]	36.07	35.35	39.64	38.83	37.50	36.76
IRI [9]	36.62	35.79	40.27	39.72	38.08	37.36
MRI [8]	36.35	35.36	40.59	39.83	38.04	37.16
PRO1	36.81	35.97	**40.79**	40.11	38.39	37.63
PRO2	**36.85**	**36.02**	40.78	**40.19**	**38.43**	**37.65**

more accurate prediction for demosaicking.

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